Cryptography

Course Overview – Second Semester 2026

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Course Overview

Course Description

Cryptography

This course introduces the principles, algorithms, protocols, and security foundations of modern cryptography.

- Mathematical foundations
- Symmetric and public-key cryptography
- Authentication and integrity
- Key establishment and management
- Cryptographic attacks and security analysis
- Practical implementation
- Emerging cryptographic technologies

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Class Schedule

Lectures

- **Monday:** 07:00–08:59
- **Wednesday:** 16:00–18:59

Tutoring

- **Monday:** 13:00–15:00

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Learning Outcomes

Learning Outcomes

By the end of the course, students will be able to:

- Explain the mathematical foundations of modern cryptography.
- Analyze symmetric and asymmetric cryptographic algorithms.
- Apply cryptographic mechanisms for confidentiality, integrity, authentication, and secure communication.
- Implement and experimentally evaluate cryptographic mechanisms.
- Assess vulnerabilities, attacks, and implementation risks.

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Course Content

Course Structure

The course is organized into four units:

1. **Cryptographic Foundations and Symmetric Cryptography**
2. **Public-Key Cryptography and Key Establishment**
3. **Authentication and Cryptographic Services**
4. **Applied and Emerging Cryptography**

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Unit 1: Cryptographic Foundations

Foundations and Symmetric Cryptography

Core concepts for the design, implementation, and analysis of cryptographic systems.

- Introduction to Cryptography and Security
- Classical Ciphers and Cryptanalysis
- Modular Arithmetic
- Shift and Affine Ciphers
- Symmetric Cryptography
- Stream Ciphers and Basic Attacks
- Block Ciphers and AES
- AES Modes of Operation

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Unit 2: Public-Key Cryptography

Public-Key Cryptography and Key Establishment

Asymmetric mechanisms for encryption and secure key establishment.

- Introduction to Public-Key Cryptography
- Number Theory and Modular Inverses
- RSA Cryptosystem
- RSA Implementation and Security
- Diffie–Hellman Key Exchange
- Key Establishment and Security Attacks

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Unit 3: Authentication and Cryptographic Services

Authentication and Integrity

Mechanisms for integrity, authentication, and digital signatures.

- Cryptographic Hash Functions
- SHA-2 and SHA-3
- Message Authentication Codes
- HMAC
- Digital Signatures
- RSA and ECDSA Signatures
- Elliptic Curve Cryptography
- ECC and Secure Key Exchange

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Unit 4: Applied and Emerging Cryptography

Modern Cryptographic Systems

Integration of cryptographic primitives into secure systems and analysis of emerging threats.

- Hybrid Cryptosystems
- Secure Communication Architecture
- Key Management
- Attacks and Security Testing
- Post-Quantum Cryptography
- Quantum Threats to Current Cryptosystems

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Teaching Methodology

Teaching Methodology

The course combines theoretical, computational, and experimental learning.

- Lectures and conceptual discussion
- Mathematical problem solving
- Cryptographic implementation
- Laboratory experimentation
- Security and attack analysis
- Project-based learning

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Technical Tools

- **Python and Cryptographic Libraries**
  Implementation and experimental analysis.
- **JupyterLab / IDEs**
  Development, testing, visualization, and documentation.
- **Git and GitHub/GitLab**
  Version control and collaborative development.
- **Network Analysis Tools**
  Protocol, traffic, and attack analysis.

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Labs and Projects

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Laboratories

Learning by Experimentation

Laboratories connect cryptographic theory with implementation and security analysis.

- 4 individual laboratory activities
- Algorithm implementation
- Experimental validation
- Attack and security analysis
- Correctness and performance evaluation

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Course Projects

Project-Based Learning

Two projects integrate implementation, experimentation, and security analysis.

1. **Project 1 – Secure IoT Communication**
   Design and implement a secure wireless communication system.

2. **Project 2 – Research Project**
   Investigate, implement, or experimentally evaluate a problem in modern cryptography.

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Project 1: Secure IoT Communication

Objective

Design and implement secure wireless text communication between IoT devices using modern cryptography.

- Confidentiality
- Integrity and message authentication
- Secure session-key establishment
- Device authentication
- Replay protection
- Detection of modified or forged messages

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Project 1: Technologies

Hardware

ESP32 or equivalent IoT devices.

Communication

ESP-NOW, Wi-Fi, or LoRa.

Cryptography

AES-GCM, ECDH, HKDF, nonces, and sequence counters.

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Project 2: Research Project

Objective

Analyze, implement, or experimentally evaluate a relevant problem in modern cryptography.

- Define a clear research problem.
- Review relevant scientific literature.
- Apply an appropriate methodology.
- Produce reproducible results.
- Evaluate security or performance.
- Present technically supported conclusions.

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Project 2: Research Approaches

Experimental Analysis

Evaluate a cryptographic algorithm, protocol, implementation, attack, or security mechanism.

Design and Implementation

Develop or improve a cryptographic solution and evaluate its correctness, security, and performance.

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Project 2: Deliverables

1. **Research Proposal**
   October 12, 2026
   - Maximum 3 pages
   - Problem and motivation
   - Objectives
   - Methodology
   - Evaluation plan

2. **Conference-Style Paper**
   December 2, 2026
   - Methodology and implementation
   - Experimental results
   - Discussion and conclusions

3. **Technical Presentation**
   December 2, 2026

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Project 2: Research Areas

- **Post-Quantum Cryptography**
- **Cryptographic Implementation Security**
- **Lightweight Cryptography and Secure IoT**
- **Cryptanalysis and Machine Learning**
- **Privacy-Preserving Cryptography**
- **Zero-Knowledge Proofs and Secure Protocols**

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Evaluation

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Evaluation Structure

First Term – 40%

- Projects / Formative Evaluation: **10%**
- Laboratories and Quizzes: **10%**
- Midterm Exam: **20%**

Second Term – 60%

- Projects / Formative Evaluation: **20%**
- Laboratories and Quizzes: **10%**
- Final Exam: **30%**

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Final Course Grade

Assessment

$$\boxed{\text{Final Grade} = 0.20\,ME + 0.20\,LQ + 0.30\,P + 0.30\,FE}$$

- **ME – Midterm Exam:** 20%
- **LQ – Laboratories and Quizzes:** 20%
- **P – Projects:** 30%
- **FE – Final Exam:** 30%

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Course Rules

Submission Policy

- Submit all activities through **Moodle**.
- Deadline: **11:59 PM** on the due date.
- Late penalty: **2 points per day**.
- More than **3 days late**: not accepted.
- Laboratories are individual.
- Projects are completed in teams of two.

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Academic Integrity

Academic Conduct

Students must maintain academic integrity in all course activities.

Prohibited practices include:

- Plagiarism
- Cheating
- Unauthorized collaboration
- Falsification of results
- Submission of work produced by others
- Unauthorized or undisclosed use of AI

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Use of Artificial Intelligence

Responsible Use

AI tools may be used only when explicitly permitted and properly disclosed.

- Students remain responsible for submitted work.
- AI-assisted content must be disclosed.
- AI is prohibited in exams.
- AI is prohibited in laboratories unless authorized.
- Unauthorized use is subject to academic integrity rules.

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Attendance

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Attendance Policy

Minimum Attendance

Students must attend at least **70%** of scheduled academic activities.

- Attendance includes lectures, laboratories, and evaluations.
- Absences must be justified according to University regulations.
- Make-up activities require an accepted justification.

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Key Academic Dates

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Key Academic Dates

Assessments and Projects

Activity	**Date**
Quiz 1	September 14, 2026
Quiz 2 / Project 1	October 5, 2026
Midterm Exam	October 7, 2026
Project 2 – Research Proposal	October 12, 2026
Quiz 3	November 9, 2026
Quiz 4	December 2, 2026
Project 2 – Paper Submission	December 2, 2026
Project 2 – Presentation	December 2, 2026
Final Exam	December 9, 2026

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Yachay Tech University

August 19, 2026

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